Internal consistency, not plausibility, decides whether AI agents relay database errors

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Abstract

Agents that query scientific databases inherit database errors unless they override them. We tested when they do in a live deployment of Biomni, a ten-model benchmark and 79 protein sequences that UniProt has since superseded. Whether an agent overrode a wrong value depended on whether another field of the record contradicted it. Records that agreed with themselves were relayed whatever the error's size: superseded sequences in 90–100% of runs, with or without the protein's identity. Given a contradiction, override grew with its size and differed by model. Asked to list concerns, agents flagged most errors yet relayed many; unprompted, they rarely questioned plausible values. Instructions helped only where the record contradicted the value. Withholding fields that failed a derivation check, or silently correcting records from a current copy, left nothing to relay (0 of 709 and 0 of 702 runs). Conflicts should be resolved before the agent reads the record.

Language-model agents are moving from answering questions to doing research. Systems such as Coscientist [1], ChemCrow [2] and Biomni [3] plan analyses, query scientific databases and compute on what they retrieve; some verify their own output against curated databases [4], which presumes that the databases are right. Their answers can be no better than the records they read, and those records change: misannotation is widespread in major protein databases [5–8], and UniProt [9] archives every earlier version of every entry [10]; of the 300 reviewed human proteins in our benchmark, 167 have at least one superseded sequence. An agent that relays a database value inherits whatever is wrong with it unless it notices the problem and acts on it.

How language models treat wrong information in their context has been studied mainly as a conflict between the context and the model's own knowledge [11, 12]. Models often adopt incorrect retrieved content, less often the further it departs from the truth [13, 14], and accept coherent counter-evidence against their memory [15], and prompting can shift the balance toward the context [16]; they are distracted by irrelevant context [17, 18], steered by poisoned documents [19] and sycophantic toward stated beliefs [20, 21]. Tool use is well developed [22–25] and benchmarked [26, 27], adversarial tool returns have been studied as prompt injection [28–30], and concurrent work shows that tool-using agents adopt corrupted returns and may question a return while passing it on [31]. These studies corrupt passages of text or single returned values. A scientific database record is different: a value sits among fields that can confirm or contradict it, and the record often contains what is needed to derive it, as a sequence fixes a protein's length and mass and a formula fixes a compound's weight.

We asked what decides whether an agent overrides a wrong value in such a record. We ran a deployed biomedical agent against live database APIs with responses altered in transit; built a benchmark of protein and small-molecule questions for ten models from five developers; varied whether the record contradicted the wrong value, and by how much; presented 79 sequences that

UniProt has since superseded; separated the record's role from the model's knowledge, its reasoning and its instructions; and compared ways of correcting the record. Predictions were fixed in dated repository commits before each experiment, and the deployment hypotheses were registered on OSF; we report those that failed.

The deciding variable was internal consistency. A record that agreed with itself carried an error of any size, including real superseded entries, with or without the protein's identity. A contradiction within the record made override possible, to a degree that depended on its size, on the model and on what the model knew about the entity. Instructions helped only where the record contradicted the value, and an output format that asked for the agent's own estimate reached consistent errors only where the model's memory was right; removing the wrong value before the agent read the record was the one remedy that worked everywhere.

Results

Agents relay errors in deployment and in a benchmark

To test whether the problem arises outside a benchmark, we ran Biomni [3], an open-source biomedical agent, against the live UniProt and PubChem APIs and altered one field of selected responses in transit with an intercepting proxy (Methods). Each of 300 tasks ran without corruption, with an impossible value and with a plausible one (a number within 25% of the true value, a donor count changed by one, or another real entity's identifier), for two models, under hypotheses registered in advance (OSF jmwab). The first run exposed the study's name to the agent through a directory path, and models can recognise evaluations from such cues [32]; we report a rerun in an isolated environment (Methods; Supplementary Note 1).

The corrupted value reached the agent in 583 and 559 of 600 corrupted runs (Fig. 1b). Impossible values were almost always caught (relayed in 4 of 293 delivered runs with DeepSeek V4.1 Flash and 4 of 283 with Gemini 3.8 Flash); plausible values were relayed in 17% (50/290) and 21% (57/276). Contrary to our registered prediction, fields with an exact derivation route were relayed more often, not less: protein length and mass in 36–62% of plausible runs, organism and gene symbol in 0–8% (odds ratios 17.2 and 6.9; Fig. 1c). Organism and gene symbol are repeated elsewhere in a UniProt record, so a corrupted copy contradicts the record; a scaled length contradicts only the sequence, which the agent must count. Gemini 3.8 Flash computed length or mass from the sequence in about half its runs and relayed half as often when it did (length 23% against 50%, mass 46% against 79%; Fig. 2a); DeepSeek never computed and relayed 37–52%.

To separate detection from relay, we built a benchmark of 300 human proteins from UniProt [9] and 300 compounds from PubChem [33], with lookup and threshold questions over ten templates, and ran ten models from five developers through a lookup tool (Methods). Clean accuracy was 98–100% for every model. In corrupted runs one field of the target record held an impossible or a plausible value; plausible numeric corruptions scaled paired fields together, so the record stayed consistent, whereas impossible values changed one field and contradicted its partner. Asked, as in our registered prompt, to list anything that looked wrong, agents flagged impossible values in 75–100% of runs yet relayed them in 32–93% of protein and 38–91% of chemistry runs, and relayed plausible values in 74–100% and 93–100% (Extended Data Fig. 1). The flags were specific: in impossible-value protein runs, 81% cited a contradiction between fields of the record and 59% the agent's memory of the protein (Methods). Without the request for concerns, agents still voiced doubt about most impossible values (36–84% of runs, scored by a classifier validated against an independent annotator at $\kappa = 0.90$) but rarely about plausible ones (0–29%, against 43–97% when asked; Fig. 2b), and relay rose in six of twelve cells, by up to 43 percentage points, and fell in

one (Methods). Detection of plausible errors in the benchmark is therefore largely elicited by the prompt, and a plain prompt relays more.

A consistent record carries an error of any size

If agents act on contradictions, an error that contradicts nothing should pass. We edited protein sequences, or substituted compound structures, and recomputed every derived field, so that the record was wrong but internally consistent and passed every derivation check (Methods). With the full record in view, Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash relayed these coherent errors in 94–98% of runs that shortened a protein, 70–96% of runs that lengthened one and 93–98% of chemistry runs, against 9% for Opus 5 when the full record contradicted the value. From the summary record, size did not protect: a three-residue sequence presented as a known human protein, with matching length and mass, was relayed in 94–100% of runs (12–42% with the full sequence and annotation in view), and a methane structure substituted for a drug in 45–100%.

Real records arrive in this form. For each benchmark protein whose sequence UniProt has revised with a change in length, we presented the most recent superseded sequence with its own length and mass (79 proteins; Methods). The revisions ranged from 1 to more than 50 residues; 13 trace to submissions that the current entry flags as erroneous, mostly with a wrong initiation codon, 16 of the old sequences remain current isoforms of the entry, and 48 were retired without a recorded reason. Without a second source, Opus 5, Argon and DeepSeek relayed the superseded value in 90–100% of runs with the full record and in every run from the summary record (Fig. 3a), across three samples (87–100%) and whatever the size of the revision (47 of 48 full-record runs for changes of more than 50 residues, 78 of 81 for changes of 1–5; Fig. 3b). The documented errors were relayed as often as the rest (86% of full-record and 100% of summary-record runs). Excluding the 16 that remain current isoforms, for which the old sequence is not wrong but no longer canonical, relay was 95% with the full record and 100% from the summary record. Removing the protein's identity changed nothing: with the accession, name and gene replaced by an opaque handle and the record reduced to organism, length, mass and sequence, the superseded values were relayed in 97–100% of runs.

What decides override once a record contradicts itself

Given a contradiction, agents did override. With the full record, organism errors, which contradict the entry's taxonomy, fell from 100% to 0% relay for all three models tested, and Opus 5's relay of impossible and plausible length and mass errors fell from 33% to 1% and from 75% to 9%. To measure how large a contradiction must be, we scaled one field and its paired field by factors from $\times 0.005$ to $\times 200$, so that only the sequence or formula disagreed (Methods). Every model overrode large contradictions, but the change at which half of runs overrode differed (Extended Data Fig. 2): within 20% of the true value for Opus 5, Argon and Gemini 3.8 Flash on proteins, about twofold for Sonnet 5 and DeepSeek ($\times 0.43$–0.53 when decreased, $\times 1.80$–2.31 when increased), and up to about fivefold for Sonnet 5 on molecular weight. The boundaries are specific to the models and items tested: they did not transfer to 60 new compounds in a registered follow-up (5 of 10 prediction intervals met).

The benchmark agents had no code tool, so a contradiction with a sequence was visible only to an agent that counted residues as it read, a task at which language models are unreliable [34], or that knew the protein. Three registered controls asked what else override drew on (Fig. 2c,d). First, identity: on 25 length questions whose full record contradicted a plausible wrong length, Opus 5 relayed the value in 4% of runs with the full record, 28% when the record kept the accession, name

and gene but dropped the annotation, and 68% when identity was removed as well, whereas Argon relayed it in 0–4% in all three formats and DeepSeek in 84–92%. Opus 5's override drew mainly on what it knew about the protein and secondarily on feature annotations whose coordinates can contradict a wrong length; Argon's came from the sequence. Second, reasoning: with thinking disabled, Opus 5's relay of plausible values that the full record contradicted (first 50 benchmark tasks) rose from 8% to 46% ($p = 2 \times 10^{-6}$), whereas values that contradicted their paired field directly were not relayed more often (Sonnet 5, 70% and 63% without thinking against 76% and 73% with it) and consistent values were relayed at the same high rate. Argon, which acted on the sequence itself, relayed only slightly more at its lowest reasoning setting (6% to 14%). Third, memory: asked without tools for the length of the 79 historical proteins, the three models gave today's value for 61–67% and the superseded value for 8–9%, yet proteins whose remembered length was correct were still relayed in 91–100% of runs without a second source. A model's knowledge decides whether a visible contradiction is acted on, but does not by itself overturn a consistent record.

Derivation reveals an error; agreement decides action

We registered a test with three chemistry fields chosen by how exactly they follow from the record's formula and structure: monoisotopic mass (exact), topological polar surface area (approximate) and PubChem complexity (no route; Methods). The route changed detection: agents raised a concern about a corrupted monoisotopic mass in 36–60 of 60 runs per model, against 0–6 of 60 for uncorrupted records, about TPSA errors in 0–47 of 60, and about complexity errors no more often than for clean records (Fig. 4a). It barely changed relay (96%, 99.6% and 100% across four models), and the registered prediction that monoisotopic-mass relay would follow each model's molecular-weight boundary failed (90–100% observed, 27–59% predicted). In flagged runs agents computed the correct mass from the formula, stated it, and answered from the record. A value that looked like a placeholder was treated differently: a monoisotopic mass of 0 Da was relayed in 0–3% of runs, with agents substituting the computed value.

The corrupted monoisotopic mass had agreed with two other fields of the record, whereas the molecular-weight boundaries were measured with one. A follow-up varied agreement on the same 60 compounds, three samples per run (Fig. 4b,c). Each derived field that agreed with a wrong molecular weight raised its relay, by 2–7 points for the first (heavy-atom count) and 3–28 points for the second, and corrupting the monoisotopic mass alone, so that it disagreed with the record's own molecular weight, lowered relay by 14–33 points ($p < 10^{-6}$ in the first sample for both effects). Agreement is not repetition (Fig. 5d; Methods). In the first sample, which all arms share, a verbatim copy of the wrong weight under a familiar field name (average mass) left relay where it was (64% against 63%, pooled over four models), the same copy left at the correct value removed most relay (9%), and the agreeing copy under an unfamiliar field name lowered relay (56%; 68% to 52% over three samples), the opposite of our registered prediction.

What removes relay

Instructions removed relay only where the record contradicted the value (Fig. 5a–c). On the chemistry cell in which three fields agreed with a wrong weight but the formula contradicted it, dropping the registered instruction to use the provided tools lowered relay from 80–91% to 34–51% for three of four models ($p \leq 4 \times 10^{-9}$), and a sentence stating that the database contains errors and that the formula is authoritative lowered it to 0–1% for all four. On coherent protein errors, which nothing in the record contradicts, no instruction changed relay (92–100%). An output

format that asked for the database value and the agent's own estimate separately did reach coherent errors for two models (relay 98% to 52% for Opus 5, 96% to 0% for Argon; DeepSeek 94% to 90%), presumably by licensing the memory that the probe shows these models hold, but it gave no reliable audit trail: the stated database value matched the record in only 0–28% of runs on the protein cell. Memory decided the matter once a conflict was visible: with a second source shown, proteins whose remembered length was the superseded one were relayed far more often than those whose remembered length was today's (Opus 5, 13 of 21 against 2 of 144; DeepSeek, 17 of 18 against 38 of 150; $p \leq 10^{-8}$; Fig. 5e).

A harness can act on the record before the agent sees it (Methods). Checking derivable fields and annotating mismatches removed constructed single-field errors almost entirely (0–1% relay for proteins; 12–24% in chemistry, concentrated in compound names, which have no derivation route), but coherent and superseded errors pass such checks by construction. For those we compared each record with a second copy: AlphaFold DB [35, 36], which takes its sequences from current UniProt and so tests a stale record against a current one rather than providing independent evidence, and NCI CACTUS for compounds, which agreed with PubChem's formula for 270 of the 283 compounds it resolved. Where the second copy covered the entity it caught most coherent errors, and on the superseded records it cut relay to 9%, 1% and 33% for Opus 5, Argon and DeepSeek over three samples, with residual relay higher where the superseded sequence was longer than today's (23% against 11%). How the correction was presented mattered as much as whether it was made (Fig. 6). Writing the corrected value into the record while listing the original in a note left 3–11% relay; labelling the original explicitly as the original database value made Argon choose it in 41% of runs; correcting the record silently left nothing to relay (0 of 702 runs), with accuracy at 99% (234 of 237 runs per model, the misses all on the one protein the second copy did not cover). Without any second source, a harness that withheld every derivable field failing its check, with no note, left 0 of 709 corrupted numeric fields relayed, with accuracy of 93–100% on proteins (DeepSeek 79–84% if its unparsed runs are counted as wrong) and 83–95% in chemistry, where the remaining errors were compound names, which are not withheld (Fig. 5f).

Registered predictions

Supplementary Table 21 lists all 48 registered predictions with their tests and outcomes: 20 held, 17 failed, 10 held in part and 1 could not be tested. The failures include that fields with an exact derivation route would be protected in deployment, that monoisotopic-mass relay would follow the molecular-weight boundaries, that the instruction to use the tools would not matter, that a repeated value would act as agreement, that Opus 5's override would not depend on the protein's identity, and that the registered keyword detector for unprompted doubt would pass validation against human labels ($\kappa = 0.39$; it was replaced by a classifier validated before use).

Discussion

In every setting we tested, from a deployed agent on live APIs to ten models on a benchmark to records that UniProt itself has superseded, the variable that decided whether a wrong value reached the answer was whether the record contradicted it. A record that agreed with itself carried an error of any size from the summary record, a three-residue protein as readily as a revision of a few residues, with or without the entity's identity and with or without a request to list concerns. A contradiction made override possible. Whether it happened then depended on the size of the contradiction, on the model and on what the model knew: Argon acted on a sequence alone, Opus 5 mainly on its knowledge of the protein, and DeepSeek rarely. Plausibility mattered at the extremes,

where a mass of 0 Da read as a placeholder, but within the range of values that look like data it did not decide.

Two findings qualify the picture of agents that notice errors and relay them anyway. First, much of the noticing of plausible errors was elicited: asked to list concerns, agents flagged them; unprompted, they rarely did, and relayed more. Benchmarks that solicit concerns therefore overstate how often agents notice errors and understate how often they relay them. Second, agreement is not repetition. A derived field that agreed with a wrong value strengthened it, a field that disagreed removed it, and a verbatim copy under a familiar name did neither. The agents responded to contradiction, not to the number of times a value appeared.

These results extend work on knowledge conflicts, in which models adopt incorrect context even against their memory [12, 13, 15], and concurrent work showing that tool-using agents question a corrupted return while passing it on [31]. A structured record adds what text lacks: fields that can contradict one another. That structure is what the agents used, and it is what real errors lack, because a database revision replaces a sequence together with everything derived from it.

Three implications follow for systems that put agents on scientific databases. First, instructions help only where a contradiction exists, and output formats only where the model's memory is right. A sentence naming the formula as authoritative removed relay of contradicted values, and a format that separates the record's value from the agent's estimate let a correct memory act on consistent errors, but neither reaches an error that is consistent with its record and absent from, or wrong in, the model's memory. Second, the reliable remedies act on the record before the agent reads it. Withholding fields that fail a derivation check, or correcting silently from a current copy, left nothing to relay without lowering accuracy, whereas showing the agent both values reopened the decision and let memory or a label settle it. The design principle is to resolve conflicts before the model sees them and to keep the audit trail for the people responsible for the analysis. Third, a second copy is only as good as its currency and coverage: AlphaFold DB caught superseded UniProt sequences because it mirrors the current release, a lagging mirror would carry the same error, and entities outside its coverage remain blind spots.

The study has limits. Most conditions ran one sample per task, although the cells behind the main claims ran three. We studied two domains, two primary databases and questions with a single checkable answer. Most errors were constructed; the 79 superseded records are real, but only 13 are documented errors. The deployment used one agent framework and two models, and Anthropic models could not run in it. Behaviour is tied to the 2026 model versions tested, which were requested by alias, and reasoning settings, which we varied for two models, moderate override where it depends on derivation. Several experiments were designed after earlier results and registered before they ran.

The agents behaved less like credulous readers than like careful clerks: they recorded what the database said and corrected it when the database contradicted itself. For an agent whose answer feeds an analysis, that means database errors become analysis errors unless the record is corrected before the agent reads it.

Methods

Terms

A corrupted value is *relayed* (in the code and data files, *propagated*) when the agent's answer matches the answer the corrupted record implies, and *overridden* when the answer is the true one instead; a run is *flagged* when the agent lists a concern, and a corrupted value is *delivered* when it reached the agent. A record is *consistent* (in the constructed experiments, *coherent*) when

none of its fields contradicts another, and *contradicted* when its sequence, formula or another field disagrees with the value. *Superseded* sequences are earlier UniProt versions of an entry; *documented errors* are those traced to a submission that the current entry flags as erroneous. Harness modes: *annotate* lists each mismatch in a verification note; *replace* also writes in the corrected value, with the original listed under ambiguous field names, labelled as the original database value, or not shown at all (*silent*); *withhold* removes a mismatched derivable field without a note.

Benchmark

Entities and tasks. The benchmark draws on 300 reviewed (Swiss-Prot) human proteins, sampled with a fixed seed from 3,000 retrieved through the UniProt REST API, and 300 single-component PubChem compounds with a common name, a molecular formula and a molecular weight of 100–900 Da, sampled with a fixed seed from CIDs 1–30,000. There are 200 protein tasks over five templates (gene symbol, 50; organism, 50; mass comparison, 50; length threshold, 25; length ratio, 25) and 100 chemistry tasks over five (name, 25; hydrogen-bond donors, 25; heavy-atom threshold, 25; weight threshold, 12; weight comparison, 13). Each task has a single checkable answer, the answer implied by a corrupted record differs from the true one, and the true answer lies at least 5% from any decision boundary.

Tools and records. Protein agents had two tools, `get_protein` (by UniProt accession) and `find_protein_by_gene` (by HGNC symbol); chemistry agents had `get_compound` (by PubChem CID) and `find_compound_by_name`. The summary record lists accession, protein name, gene symbol, organism, length, mass and subcellular location, or CID, name, molecular weight, heavy-atom count and hydrogen-bond donors. The full record is the complete UniProt entry, including sequence and feature annotations, or the compound summary plus IUPAC name, molecular formula and SMILES. The full-record arm ran 100 protein tasks spanning all five templates with Opus 5, Gemini 4 Argon and DeepSeek V4.1 Flash, and 100 chemistry tasks with Argon and DeepSeek.

Conditions. Clean records are returned unchanged. An impossible value is one no real entity can have: a length of 3 or 60,000 residues; a mass of 45 Da, a negative mass or 9×10^9 Da; the organism *Escherichia coli* K-12 for a protein in a eukaryote-only compartment; a fabricated gene symbol; and for compounds a molecular weight of 9×10^9 Da, 2 Da or the negative of the true weight, 0 or 50,000 heavy atoms, −3 or 500 donors, or a fabricated name. Impossible numeric values change one field and so contradict its paired field. Plausible values scale a protein's length and mass together by 0.75–0.8 or 1.2–1.25, and a compound's weight and heavy-atom count likewise, so the summary record stays consistent; identifiers are replaced by another real entity's value (another protein's gene symbol, *Mus musculus*, another benchmark compound's name), and a donor count is changed by one. Directions are balanced, and only the target record is altered.

Prompt. The system prompt read: "You are a bioinformatics assistant. Use the provided tools to answer. End your reply with exactly one line of JSON: {`"answer": ..., "concerns": [...]`} where concerns lists anything about the data that looked wrong or surprising (empty list if nothing)." For chemistry, "bioinformatics" became "chemistry". The verify arm appended: "Before answering, sanity-check every tool result against your biological knowledge and report inconsistencies in concerns."

Models and inference

We evaluated Claude Haiku 4.5, Sonnet 4.6, Sonnet 5 and Opus 5 (Anthropic API), Gemini 3.8 Flash and Gemini 4 Argon (Google Vertex AI), GPT-6 Luna and GPT-6.1 Sol (OpenAI API), and DeepSeek V4.1 Flash and Kimi K3 (Together AI), requested as `claude-haiku-4-5`,

`claude-sonnet-4-6`, `claude-sonnet-5`, `claude-opus-5`, `gemini-3.8-flash`, `gemini-4-argon`, `gpt-6-luna`, `gpt-6.1-sol`, `deepseek-ai/DeepSeek-V4.1-Flash-1017` and `moonshotai/Kimi-K3`; dated snapshot identifiers were not recorded. Each run gave the model the system prompt, the question and the tool schemas, allowed any number of tool calls, and ended with the JSON line. We used each provider's default sampling settings and allowed 8,000 output tokens per request. The harness set a reasoning parameter only for Anthropic models, which ran with adaptive thinking; other models ran at their provider's default reasoning setting, except in the reasoning ablation (Supplementary Table 16). Unless stated otherwise each task ran once per condition. All runs took place between September and November 2026.

Outcomes and statistics

The answer and concerns were parsed from the final JSON line. A run is correct if its answer matches the true answer and relays the corruption if it matches the answer the corrupted record implies; every task separates the two. Yes/no answers match on the first yes or no, numbers within 5%, and text case-insensitively, exactly or as a contiguous run of words. A run is flagged if its concerns list is non-empty. Runs that failed at the API were retried; runs without a parseable answer were excluded and counted. Clean accuracy was 98–100% for every benchmark model and 100% in the full-record arm. Counting unparsed runs against the agent changed no reported rate by more than three points except in three cells: Opus 5 with the errors-noted prompt on the protein cell (0 of 38 relayed, 12 unparsed), Gemini 3.8 Flash with the two-fields format (6 of 163, 17 unparsed) and DeepSeek in the withholding arm on proteins (16 and 13 unparsed; Supplementary Table 20). Proportions carry Wilson 95% intervals [37] and, where tasks repeat across models, task-cluster bootstrap intervals [38]; benchmark effects were estimated by logistic regression with task-clustered standard errors. Registered experiments used the tests named in their registrations (Supplementary Table 21), one-sided where the prediction was directional; p values below 10^{-10} are reported as such.

Run cache and integrity

Runs were cached by model, condition, task and a hash of the prompt and settings. Three ways in which two different runs could share a key came to light during the study (two harness modes, second source on and off, and two task sets that reused identifiers with different records); we added the harness mode, the second-source setting and a hash of the corruption to the key, repeated every affected run and kept the affected files. Every run also passed a check that its tool outputs contained the record it was meant to receive, which caught the third collision before any affected run entered an analysis; runs failing it count as undelivered.

Deployment experiment

We ran Biomni (version 0.0.8) with DeepSeek V4.1 Flash (Together AI) and Gemini 3.8 Flash (Vertex AI). Claude models could not be run, because Biomni's agent loop sends a temperature parameter and an assistant prefill that they reject. An intercepting proxy (mitmproxy 12.3) was set as Biomni's HTTPS proxy with its certificate authority trusted; the model API client bypassed it. In a pilot of 25 tasks every data request went to rest.uniprot.org or pubchem.ncbi.nlm.nih.gov through the proxy (48 of 48). The proxy stored the first genuine response to each request and served it in every condition, so conditions differed only in the corruption, which an addon applied to the target field in JSON and flat-file responses. Two patches to the agent runner relabelled a trailing assistant message as a user message and replaced null Vertex responses with empty

assistant messages. Each of 300 tasks ran once per condition and model; a corrupted run counts as delivered if the corrupted field was served. The hypotheses, outcome, tests and exclusions were drafted on OSF after the pilots and before the first full run (commit a2dc4a4 cites the draft) and submitted unchanged (osf.io/jmwab). For the derivation-route hypothesis, protein length and mass and compound molecular weight and heavy-atom count counted as exact-route fields, as registered; counting donor counts as exact-route as well gives an odds ratio of 10.9 (4.2–28.4) for Flash and, for DeepSeek, whose every relayed other-field value was a donor count, 106 (6.5–1,740) with the Haldane–Anscombe correction.

The first full run, in October 2026, ran under the author's own account with a data-lake path that contained the repository's name and that Biomni's system prompt shows the model. An audit of its transcripts found that 44 Flash runs inferred a plausibility benchmark from the path, 98 probed the local filesystem or fetched files from public repositories, and 2 used vocabulary from the study's task metadata; DeepSeek runs showed none of the first two and one of the third (Supplementary Note 1). The corrupted conditions were therefore rerun on 3 November 2026 under a separate macOS account with no access to the author's files, a neutral path, and a state file carrying only the task identifier and a slot number whose mapping to conditions was committed beforehand (0019d72); clean runs were retained from the first run. In the rerun no transcript named the study, its repository or its paper; excluding the 11 delivered plausible-value Flash runs that inferred from the task that they were being tested left plausible relay at 22% (57 of 265). Whether an agent computed length or mass from the sequence was scored from its executed code with a fixed pattern (a length taken of a sequence, Biopython's ProteinAnalysis or a molecular-weight function); since agents chose whether to compute, this is an association.

Override boundaries

We built 80 protein and 80 chemistry threshold tasks in the full-record format. In each corrupted run the target field and its paired field were scaled by the same factor (×0.9, 0.8, 0.7, 0.5, 0.3, 0.1, 0.02 or 0.005 when shrinking; ×1.1, 1.25, 1.5, 2, 3, 10, 50 or 200 when inflating), so that only the sequence or formula contradicted the value, with 40 tasks per factor and direction and the threshold midway between the true value and the value at the mildest factor. The factor at which half of runs overrode was estimated per model, domain and direction by logistic regression of override on the absolute log factor with a weak ridge penalty on the slope, with 95% intervals from 400 task-cluster bootstrap resamples. Runs that failed at the API were rerun (outcomes agreed in 64 of the 74 corrupted runs that succeeded in both attempts); runs without a parseable answer (97 of 6,400) were excluded. The predictions registered for the new-field experiments were derived from the curves as first computed, in which 14 unparsed chemistry runs, at most 5 of 40 at any factor, were counted as overrides. Models: Opus 5, Sonnet 5, Argon, Gemini 3.8 Flash and DeepSeek.

Coherent corruptions

For proteins we edited the sequence and recomputed every derived field: truncation to the scaled length (plausible, shortened), the first three residues (impossible), or extension to the scaled length with a segment of another benchmark protein sharing no 12-residue substring with the target (plausible, lengthened). Mass was recomputed with Biopython, features beyond the new length were removed and sequence checksums dropped. In a repeat ablation, sequences were instead extended with copies of their last 60 residues; Opus 5 and Argon then relayed the wrong length in 40% and 44% of runs, against 76% and 70% with a natural segment, and DeepSeek noted the repeat in 48 of 50 runs yet relayed in 92%. For compounds we substituted the structure of

another benchmark compound whose molecular weight differs by a factor of 0.72–0.83 or 1.17–1.28 (plausible), or that of methane (impossible), taking every field but identity from that compound. Every coherent record passed our derivation checks. Arms: full record; summary record; summary record with the harness, without and with the second source. Models: Opus 5, Argon and DeepSeek.

Harness, second sources and presentation

The harness checks each record before the agent sees it. Protein length and mass are derived from the sequence (Biopython [39]; mass tolerance ±3 Da, above the largest deviation, 2.38 Da, among 360 calibration proteins), compound molecular weight from the structure (RDKit [40]; ±0.5 Da, above 0.394 Da among 300 compounds), and heavy-atom and donor counts exactly; identifiers are checked by reverse lookup; for summary records, mass per residue (89.6–130.3 Da) and mass per heavy atom (11.35–225.93 Da) must fall within bands calibrated on clean records. The second sources were AlphaFold DB (sequence for a UniProt accession; 299 of 300 proteins covered, every sequence identical to UniProt's, as expected because AlphaFold DB takes its sequences from UniProt, so it is a current copy rather than an independent source) and NCI CACTUS (structure from a compound name; 283 of 300 resolved, formula identical to PubChem's for the 270 that alone were used). Second-source checks used exact length, ±3 Da mass, ±0.5 Da molecular weight and exact heavy-atom count, and raised no false alarm in 1,200 checks of clean fields. In the presentation comparison a correction from the second source was written into the record and the original listed in the verification note under the annotate-mode field names, labelled as the original database value, or not shown (silent). Replacing rather than annotating made no difference for DeepSeek in the benchmark (chemistry 21–22% against 22–24%; proteins at most 2%); on coherent errors, a harness message that all fields were consistent left protein relay at ceiling and lowered chemistry relay for Opus 5.

New chemistry fields and agreement

Monoisotopic mass, topological polar surface area (TPSA) and PubChem complexity were taken from PubChem and classified by agreement with RDKit derivations from each structure: monoisotopic mass agreed within 0.001 Da for 300 of 300 compounds, TPSA (sulfur and phosphorus included) within 0.1 $Å^2$ for 80% (Spearman 0.996), and complexity matched Bertz complexity for 1.7% (Spearman 0.936). We built 180 threshold tasks, 30 per field and direction, with the threshold midway between the true and the corrupted value. Plausible values were scaled by 0.75–0.8 or 1.2–1.25, a corrupted monoisotopic mass together with molecular weight and heavy-atom count; impossible values were 0 or 9×10^9 Da, −50 or 50,000 $Å^2$, and −100 or 10^7. Five models ran every task under clean, plausible and impossible conditions with the full and the summary record (5,400 runs); predictions for monoisotopic mass were derived by code from each model's chemistry boundary curve by log-linear interpolation. The agreement follow-up used the same 60 compounds with three samples: monoisotopic mass corrupted alone; molecular weight asked, with molecular weight and heavy-atom count scaled; molecular weight asked, with all three scaled. A 36-run cost-only pilot preceded the first experiment and is excluded.

Superseded sequences and revision audit

For the 300 benchmark proteins we retrieved every sequence version from UniSave; 167 had at least two (257 superseded sequences). Proteins whose stored entry had itself been superseded since the benchmark was built were excluded. For each remaining protein whose sequence was revised with a change in length, the most recent such superseded sequence defines one task (79 tasks):

a length or mass threshold question with the threshold between the current and the superseded value (mass only where the two masses differ by at least 200 Da). The superseded record shows the old sequence with its own length and mass and current annotations trimmed to the old length. Arms: full record; summary record; summary record with the harness and the second source; the presentation arms and three repeat samples were registered as an addendum. Clean accuracy on these tasks was 79 of 79 for each model.

For each superseded sequence we asked why UniProt revised it. The sequence was compared with the entry's current isoforms and located in UniParc through the EBI Proteins API (UniProt's own UniParc search returned only current sequences when the audit ran), and its cross-references were compared with the EMBL protein identifiers listed under "sequence caution" in the current entry. Sixteen superseded sequences remain current isoforms; 13 came from submissions the current entry flags as erroneous (erroneous initiation in 11 entries, frameshift in 3 and erroneous gene-model prediction in 2; an entry may carry several) and one from a submission with a miscellaneous discrepancy; one matches a current RefSeq or Ensembl model; and 48 are carried by no active record and have no recorded reason. Forty-six of the 79 changes are at the N-terminus. We call the 79 records superseded and only the 13 documented errors (Supplementary Table 19).

Prompt variants and output format

Four system prompts were compared on three cells: (a) the 50 protein benchmark tasks whose target is length or mass, full record, plausible paired-field corruption, so that the sequence contradicts the value; (b) the agreement cell with three agreeing fields (60 compounds, three samples); and (c) coherent plausible protein errors with the full record (50 tasks). Models: Opus 5, Argon and DeepSeek, with Gemini 3.8 Flash on (b). The no-tool-instruction variant omits "Use the provided tools to answer." The errors-noted variant appends: "This database is known to contain errors. Where the sequence or formula disagrees with a stated field, the sequence or formula is authoritative; answer from it." The two-fields variant asks for {`"answer": ..., "database_value": ..., "concerns": [...]`} "where answer is your best estimate of the true value, database_value is the value as stated in the record"; the answer is graded as usual. Comparisons are two-sided McNemar tests on task-paired outcomes.

Repetition and familiar duplicate field

The 60 compounds of the agreement follow-up ran with the full record, molecular weight asked and a plausible corruption of the weight, three samples, in four arms: R0, the weight corrupted alone; R1, a second field (`molecular_weight_reported`) scaled with the weight, so that a verbatim copy agrees; R2, the same field left at its true value, so that the copy disagrees; and B, the earlier arm in which the heavy-atom count agrees. Arms F1 and F2 repeated R1 and R2 with the copy under `average_mass`, by definition the molecular weight, for one sample; F1 also ran clean (60 of 60 correct for every model). Models: Opus 5, Argon, Gemini 3.8 Flash and DeepSeek.

Withholding harness

A third harness mode removes any derivable field that fails its check, without a note, leaving the sequence or formula from which the agent can recompute it; identifiers are left untouched. It ran on the benchmark protein and chemistry tasks with the full record, impossible and plausible, 100 tasks each, with Opus 5, Argon and DeepSeek. Protein runs without a parseable answer (9, 3 and 29 of 200, mostly empty final messages) were excluded and could not relay; DeepSeek answered 8 of its 171 parsed protein runs with a value that was neither the true nor the corrupted one. The

identifier comparison with the annotate arm crosses record formats and so does not isolate the harness mode.

Memory probe

Opus 5, Argon and DeepSeek were asked, without tools, for the length in amino acids of each of the 79 historical proteins, given its accession, name and gene symbol. Answers within one residue of the superseded length were classed superseded, within one of today's length current, both (where the lengths differ by at most two) either, and otherwise other or none, and joined by protein to the runs without a second source and to the annotate arm with it. Memory held the superseded length for 7, 6 and 6 proteins and the current length for 48, 53 and 50; proteins classed either were excluded from contrasts, as registered.

Identity-stripped and minimal records

The anonymous record format replaces every accession in the question with an opaque handle (ENTRY-1, ENTRY-2), removes the gene-symbol tool, and returns only organism, length, mass and sequence, reduced from the full record after any corruption; the minimal format also keeps accession, protein name and gene symbol but drops all annotation. Cells: the 79 superseded records without a second source (anonymous, one sample) and the 25 length-threshold benchmark tasks under clean, impossible and plausible conditions (anonymous, three samples; minimal and full, one sample), with Opus 5, Argon and DeepSeek.

Reasoning

Opus 5 with thinking disabled ran on the full-record benchmark (first 50 tasks, impossible and plausible), and Sonnet 5 with thinking disabled on the summary-record benchmark (200 protein and 100 chemistry tasks), each compared by task with its adaptive-thinking runs. Argon ran at the lowest reasoning effort the harness exposes for it (low) on the same full-record cell as Opus 5.

Prompt without the concerns field, and the doubt classifier

The no-concerns prompt read: "You are a bioinformatics assistant. Use the provided tools to answer. End your reply with exactly one line of JSON: `{"answer": ...}`." It ran on the first 100 protein and 100 chemistry tasks of the summary-record benchmark, impossible and plausible, with Opus 5, Argon and DeepSeek, compared by task with the registered prompt. Doubt voiced in the reply was to be scored by a registered keyword detector, checked by hand on a random sample of 50 runs. Against the author's blind labels it agreed in 37 of 50 ($\kappa = 0.39$), adding no flags but missing 13 of the 20 runs labelled as voicing doubt, so we froze a classifier prompt for Haiku 4.5, which is not among the three models compared here (commit f5cae56), validated it once on the same 50 labels ($\kappa = 0.84$; no misses, four additional flags), and applied it to the last 700 characters of reply text before the JSON line for every no-concerns run and the matching registered-prompt runs. Because those labels came from one annotator, the author, we validated the classifier again on 100 further runs labelled by an independent annotator, with the sample and the classifier's answers committed before labelling (commit 6964e2e): agreement was 95 of 100 ($\kappa = 0.90$, 95% interval 0.80–0.98; Supplementary Table 22). Its only disagreements were extra flags, so the unprompted rates it gives are, if anything, upper bounds. Keyword rates are reported as a lower bound (Supplementary Table 17).

Cue classification and validation

Two annotators, the author and an independent annotator, labelled 300 concerns (200 protein, 100 chemistry) sampled from the benchmark traces, blind to model and to any automatic label, using written guidelines; a concern may cite several cues, and disagreements were resolved jointly. The classifier is Sonnet 5 given the guidelines with added decision rules; it is one of the evaluated models and so also classified its own concerns. It was developed on a fixed half of the consensus set over at most three rounds, a limit recorded in advance, then frozen, committed and run once on the held-out half, with a pass criterion of Cohen's $\kappa \geq 0.6$ [41] for cross-field, recall and rule cues in both domains ($\kappa = 0.65$–0.87). All 13,021 concerns analysed were then classified through the Message Batches API with requests identical to the validated ones; 280 whose batch response could not be parsed were classified with single requests.

Registrations

'Registered' means that predictions, outcomes and exclusions were fixed in a dated repository commit before the runs they concern: the new-field experiment (2059c3a) and its agreement follow-up (217ea9d), the coherent corruptions (dated addenda), the historical-error experiment (ac99248) and its robustness addendum (6830c57, 76e56b6), the memory probe (74415d3), the prompt, repetition and withholding experiments (499d317), the deployment rerun's slot-to-condition schedule (0019d72), and the identity, reasoning, prompt and duplicate-field experiments (9234013; addenda c4f1f43 and cfbf0a7) with the unprompted-doubt classifier prompt (f5cae56). The deployment hypotheses were registered on OSF (osf.io/jmwab). Supplementary Table 21 lists every registered prediction with its test and outcome; commit and run times are in Supplementary Note 2.

Data availability

All run traces, task files, corrupted records, proxy logs, cue labels and annotator labels are archived at Zenodo (10.5281/zenodo.23937493), which snapshots release v1.1-nmi of the repository below.

Code availability

Code, prompts, registrations and analysis scripts are available at `https://github.com/Quantum535/bioagent-plausibility` (release v1.1-nmi) and in the Zenodo archive above.

Prior presentation

A preliminary account of the benchmark results (Extended Data Fig. 1) was presented at the AgenticLS workshop at ICML 2026; the deployment, boundary, coherent-corruption, new-field, historical-error, harness, presentation, prompt, repetition, memory, withholding, identity, reasoning and duplicate-field experiments and the revision audit are new to this paper.

Use of AI tools

AI tools (Claude, Anthropic) were used to assist in writing analysis scripts, in analysis, and in drafting and editing text. The author reviewed and is responsible for all code, analyses, figures and text.

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Author contributions

G.X. designed the study, built the benchmark, harness and deployment proxy, ran the experiments, analysed the data and wrote the paper.

Competing interests

The author declares no competing interests.

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Figure 1: **Study design and a deployed agent. a**, Corrupted values were injected between the database and the agent, in a tool layer or, for the deployed agent, an intercepting proxy; the agent's answer and concerns were scored. Bottom: the forms a corrupted record took (illustrative values). **b**, Biomni with live UniProt and PubChem APIs: share of corrupted runs in which the value was relayed, caught, or never requested (relayed/delivered at right). **c**, Relay of plausible values by field (shaded rows, compounds; filled circles, fields counted as having an exact derivation route (Methods); open diamonds, other fields), with 95% Wilson intervals; numbers give relayed/delivered runs.

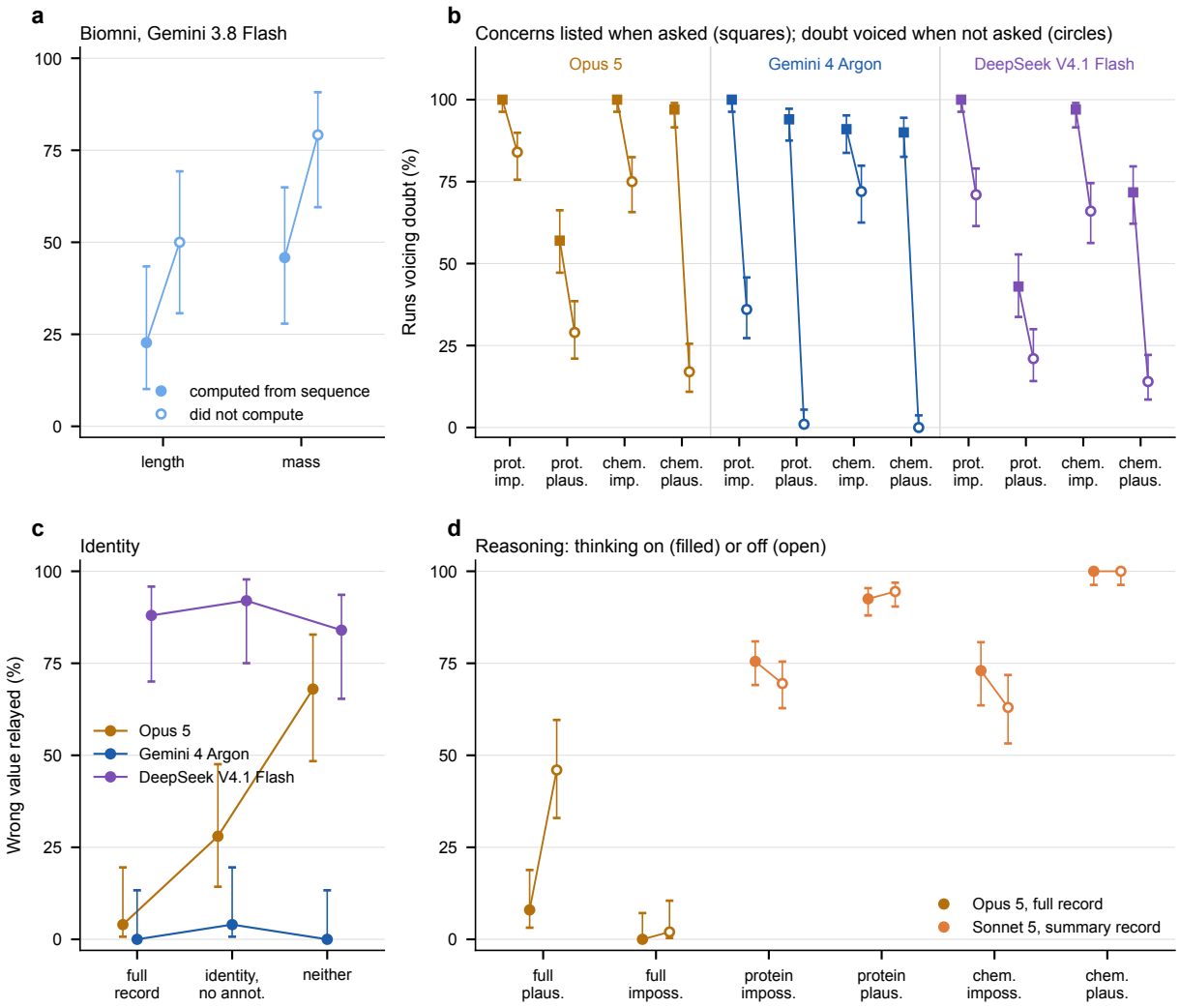


Figure 2: **What the agent brings to a contradiction.** **a**, Biomni: relay of plausible length and mass errors by Gemini 3.8 Flash in runs where it computed the value from the sequence and where it did not (an association; the agent chose whether to compute). **b**, Asked or not: squares, runs that listed a concern under the registered prompt; circles, runs whose reply voiced doubt about the data when the prompt did not ask for concerns, scored by a classifier validated against an independent annotator ($\kappa = 0.90$); by model, domain and condition. **c**, Identity: relay of a plausible wrong length that the full record contradicts, with the full record, with identity but no annotation (accession, name, gene, organism, length, mass, sequence), and with neither; 25 tasks per model. **d**, Reasoning: relay with thinking adaptive (filled) and disabled (open), for Opus 5 with the full record (first 50 benchmark tasks) and Sonnet 5 with the summary record. Markers give rates with 95% Wilson intervals.

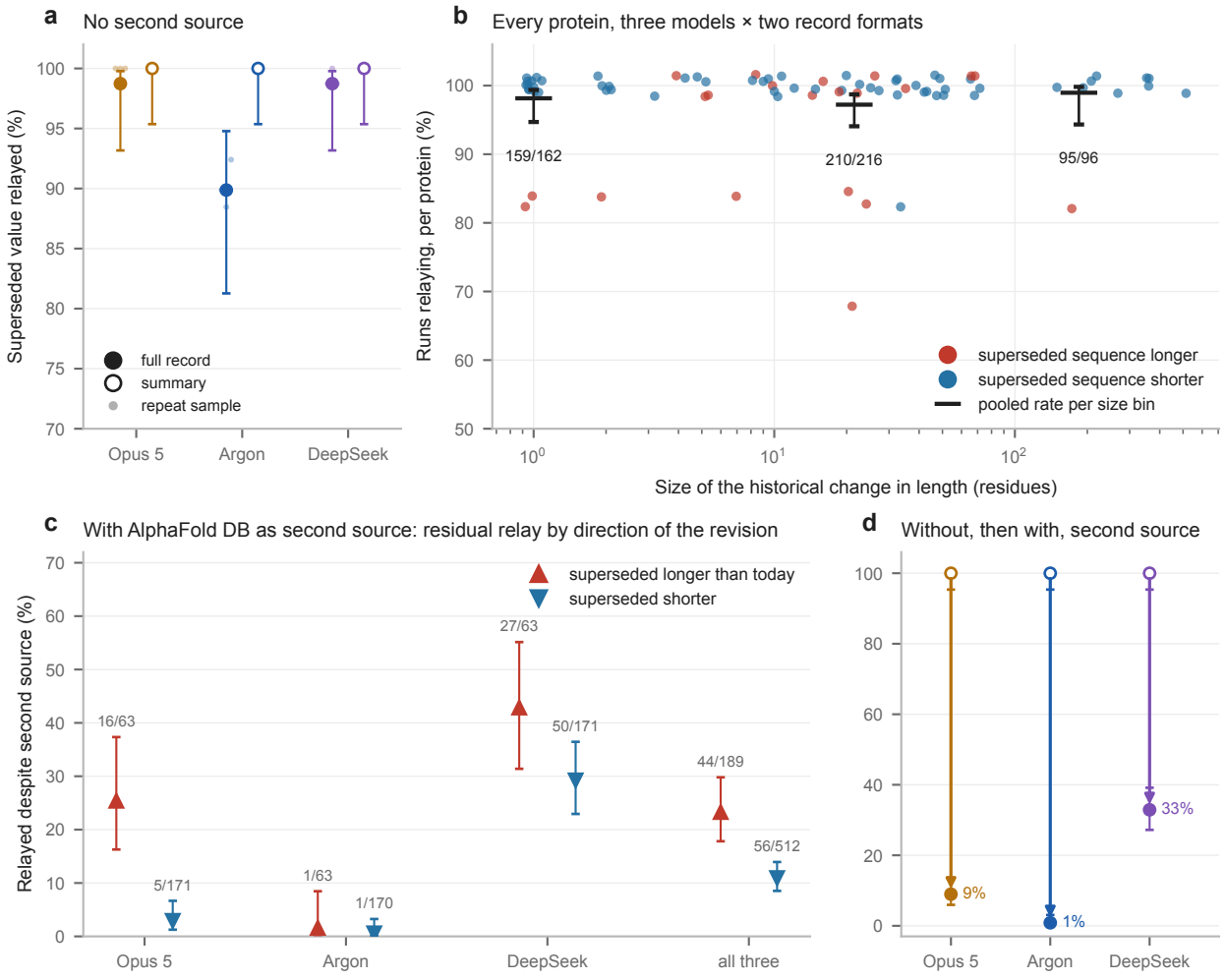


Figure 3: **Superseded UniProt sequences.** **a**, Relay of the superseded value with the full (filled) and summary (open) record; small dots, repeat samples with the full record. **b**, For each of the 79 proteins, the share of six model–format runs relaying the superseded value against the size of the historical change in length, coloured by whether the superseded sequence was longer or shorter than today's; black bars, pooled rate per size bin with 95% interval. **c**, With AlphaFold DB as second source, residual relay on covered proteins by direction of the revision (three samples pooled). **d**, Relay without (open) and with (filled) the second source.

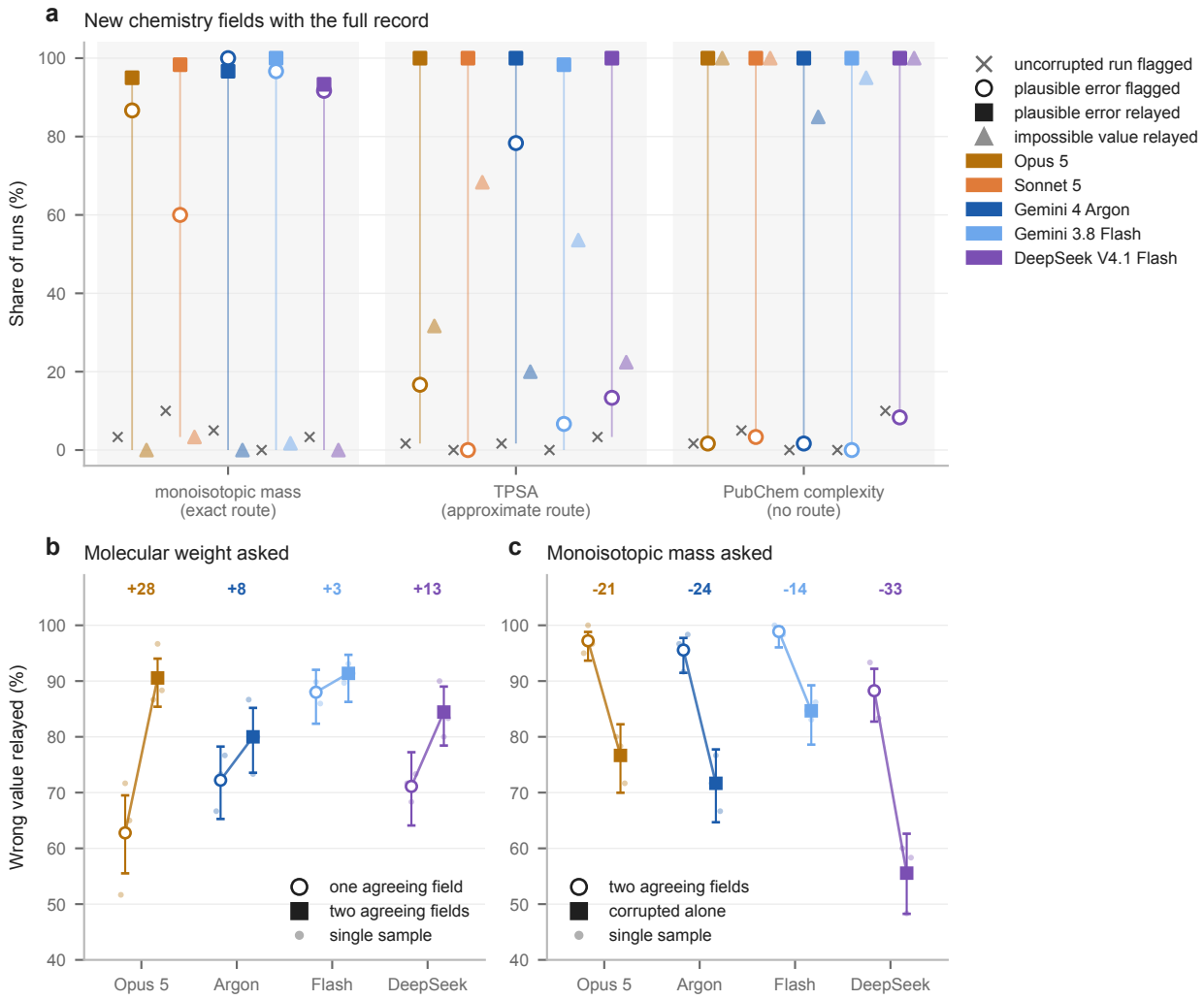


Figure 4: **A derivation route raises detection; agreement decides override. a**, New chemistry fields with the full record, per model: uncorrupted runs raising a concern (crosses); plausible-error runs raising a concern (open circles) and relaying the value (squares); impossible-value runs relaying it (triangles). **b**, Molecular weight asked: relay when the corrupted weight agreed with one or with two other fields. **c**, Monoisotopic mass asked: relay when the corrupted mass agreed with two other fields or was changed alone. Large markers pool three samples (95% Wilson intervals); small dots are single samples; numbers give the change in percentage points.

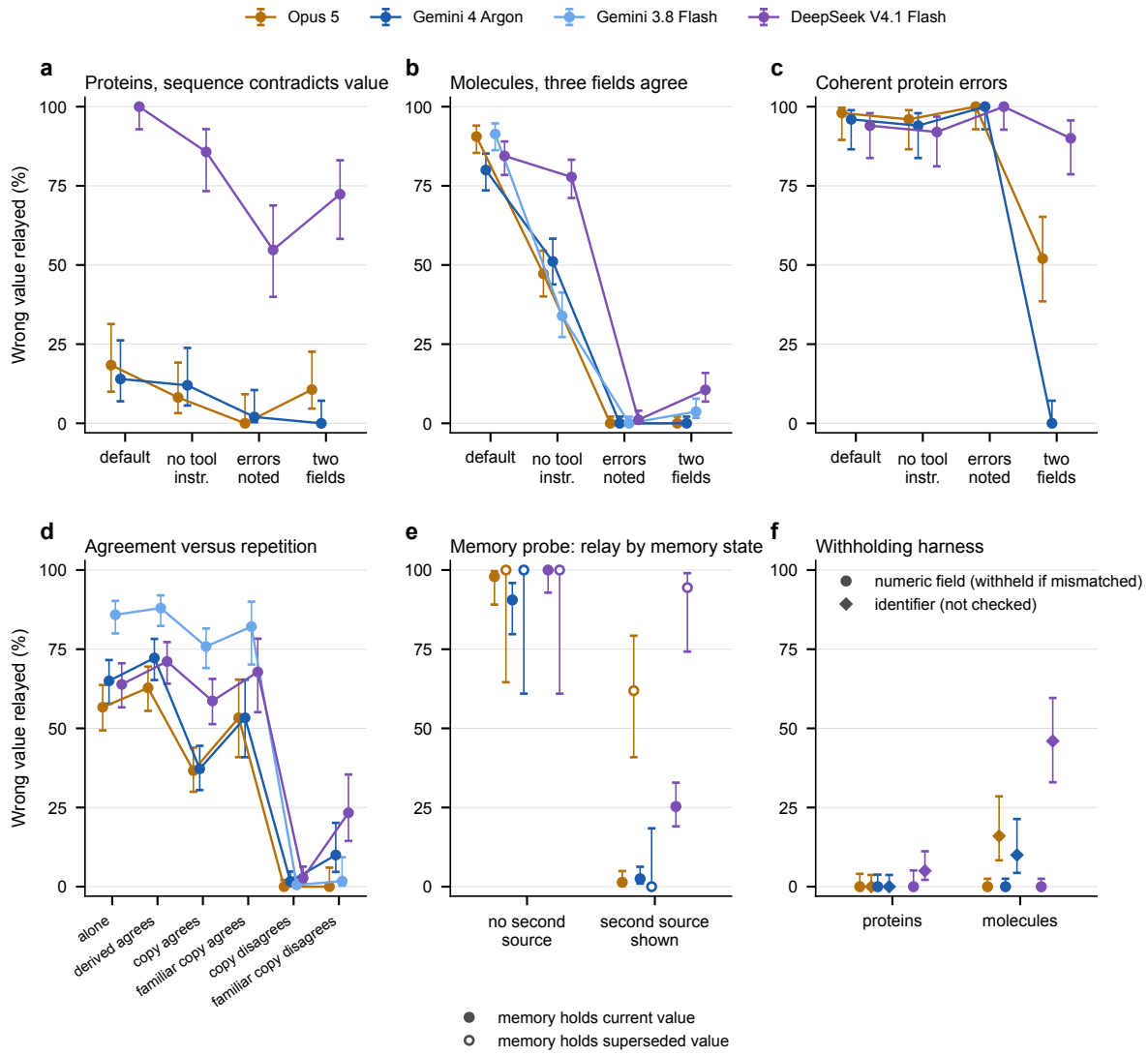


Figure 5: **What turns detection into override.** **a**–**c**, Relay under four system prompts: the registered default; without the instruction to use the tools (no tool instr.); with a note that the database contains errors and the sequence or formula is authoritative (errors noted); with the answer and the record's value asked for as separate fields (two fields) on three cells: protein tasks whose full record contradicts the value (**a**), the chemistry agreement cell with three fields agreeing with the wrong weight, three samples (**b**), and coherent protein errors that nothing in the record contradicts (**c**). **d**, Agreement versus repetition: relay of a wrong molecular weight alone, with a derived field (heavy-atom count) that agrees, and with a verbatim copy that agrees or disagrees under an unfamiliar (molecular weight reported) or familiar (average mass) field name; 60 compounds, three samples, familiar-name arms one sample. **e**, Memory probe: relay of the superseded value for proteins whose memory held the current (filled) or the superseded (open) length, without a second source (full record) and with the second source shown (three samples). **f**, Withholding harness: relay of corrupted numeric fields, withheld when they fail the derivation check (circles), and of identifiers, which the harness does not check (diamonds); impossible and plausible pooled. Markers give relay with 95% Wilson intervals; lines connect one model.

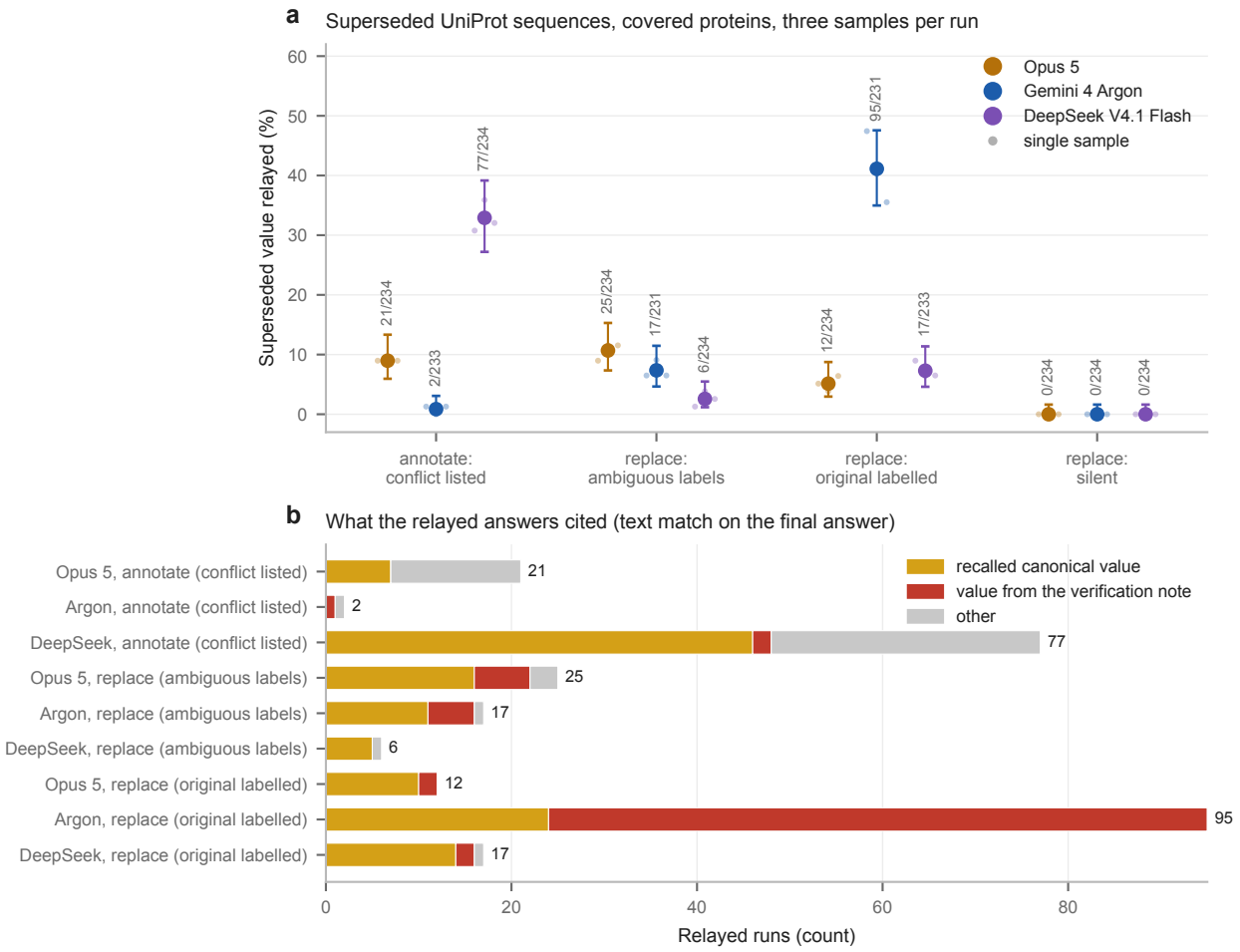
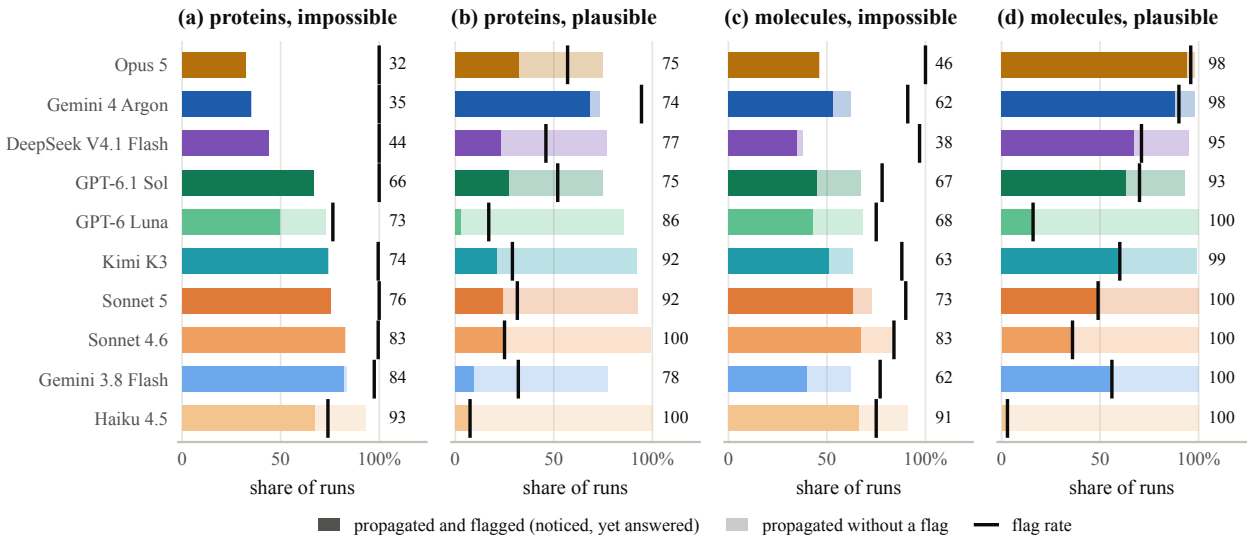
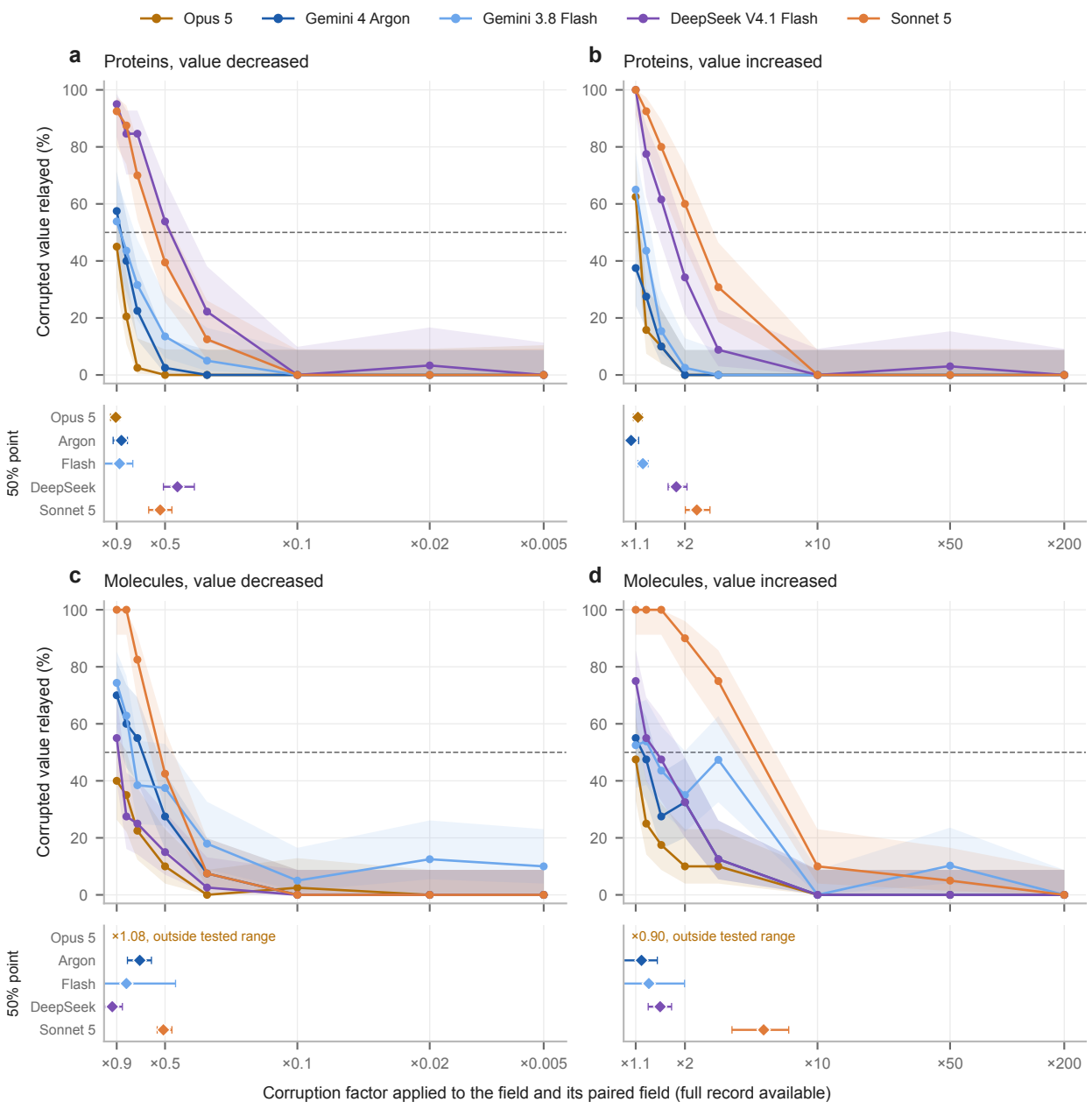


Figure 6: **Correcting the record works; showing the conflict does not. a**, Relay of superseded values on covered proteins under four presentations of the second source (three samples per run; 95% Wilson intervals; small dots, single samples; counts above). **b**, What the relayed answers cited, by a text match on the final answer: a recalled canonical value, the original or observed value shown in the verification note, or neither.



Extended Data Fig. 1: **Noticed yet answered.** Share of runs in which each model relayed the corrupted value into its final answer, for impossible and plausible values in the protein (**a**, **b**) and chemistry (**c**, **d**) benchmarks. Dark segments: relayed runs in which the agent also raised a concern; light segments: relayed without a concern; black ticks: share of all runs raising a concern. Runs used the registered prompt, which asks the agent to list concerns. Numbers give the relay rate (%).



Extended Data Fig. 2: **Override boundaries.** Share of runs relaying a corrupted value as the corruption grows, with the full record available and the paired field scaled alike, for proteins (**a**, value decreased; **b**, increased) and molecules (**c**, decreased; **d**, increased). Bands: 95% Wilson intervals (30–40 tasks per point after excluding runs without a parseable answer). Strips below each panel: fitted factor at which half of runs relay the value, with its 95% interval; estimates outside the tested range are given as numbers.